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Methodological Problems and Pitfalls in the Study of Human Metacognition

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In this chapter, we discuss several methodological issues concerning metacognitive accuracy. These issues are of importance because they could serve to qualify interpretation of experimental findings concerning the nature of and mechanisms underlying people's metacognitive abilities. First, we show how the nature of the final test itself, and, in particular, the number of alternatives in that test, influences assessed accuracy of prediction. A review of the literature is given showing that accuracy of metacognitive prediction increases along with the number of alternatives at time of second test. The reasons for this strong relation are discussed. Second, we discuss how restricted range on either the judgments themselves or on the criterion variable can influence the accuracy of metacognitive predictions. We present an experiment that illustrates the impact of this potential confound. Third, we discuss problems that may arise when comparing groups that show a different mean level of problem solving, recall, or recognition. Dissociations in metacognition, occurring among such groups, may have implications for our understanding of the architecture of cognition. But we need to be confident that the dissociations are real and not a mere consequence of the methods of measurement. The use of nonparametric as compared to parametric statistics for measuring accuracy when the level of memory performance varies radically between patient groups is discussed, as are other methods of control that have been used by various researchers. Finally, following Glenberg and his colleagues we stress the importance of informing the subject as to the nature of the upcoming

memory test prior to making judgments. Given that there is no general pervasive metacognitive knowledge that is equally useful on all tests, then subjects' lack of appreciation of the nature of the test they are about to take can lead to serious distortions in assessment of their metacognitions. Subjects may know what they know, but unless they also know what kinds of questions they will be asked they may not be able to assess how well that knowledge will serve them. This lack of knowledge about the nature of the test may occur because the test conditions are not specified, because they are incorrectly specified, or because the skill level of the subject is not adequate to understanding the task.

One would like to be able to interpret differences in the accuracy of metacognitive judgments in terms of differential cognitive processes, differences in the transparency of certain tasks to conscious inspection, or differences, perhaps, among different patient groups, in mechanisms by which judgments are made. However, such cognitive interpretation may not always be straightforward because some methodological issues may cloud the results. In the studies that we discuss here subjects make one or more predictive judgments about their own future performance on a test of memory, problem solving, or comprehension. They are then given the final test and the relation between the predictive judgments and the final test results are the metacognitive "accuracy" measure of interest. We will deal mainly with experiments that investigate micropredictive accuracy or discrimination, that is, does the subject know on which questions he or she will do well or poorly? This micropredictive accuracy is often measured by a correlation coefficient, such as the nonparametric Goodman-Kruskal gamma measure, that relates the ranking of the questions to the correctness of response on the final test. In some instances, though, we will refer to studies in which macroprediction, or calibration, is of interest. In this case the prediction is one of how good performance will be, overall, without respect to which particular questions contribute in which way to that performance. Micro- and macroprediction are both metacognitive indices, but measure aspects that may be but are not necessarily linked (see Yates, 1990, p. 57). In this first section we discuss several factors concerning the relation between the judgments and the subsequent test that, failing careful consideration and control, could produce spurious results.

Number of Test Alternatives

A simple factor that can influence the micropredictive accuracy of feeling-of-knowing (FOK) judgments, for reasons having little to do with the mechanisms underlying human metacognition, is the number of alternatives presented in the final test. Suppose the subject does not know the answer to a particular question and knows that he or she does not know, and hence gives the question a very low FOK rating. If he or she then gets that question wrong on the final test, this correspondence between the low rating and the incorrect performance will contribute to a positive correlation. This would be an appropriate demonstration of accurate metacognition. However, the chance that the person will get such a question wrong depends on the number of alternatives presented at the time of test. If the test is a two-alternative recognition test, then the odds of picking a correct answer with no knowledge are quite high. As the number of alternatives becomes large, the chances of guessing the correct answer become increasingly small. Insofar as these correct guessing responses should decrease the correlation, which is the measure of metacognitive accuracy, and insofar as more of these correct guesses are expected with fewer test alternatives, we would expect to find lower gamma correlations with few than with many alternatives.

Leonesio and Nelson (1990) noted this possibility, and hence constructed a 19-alternative forced choice test. They say: "The large number of alternatives helped to reduce the noise that necessarily occurs in a forced-choice recognition test; that is, the greater the number of distractors, the less likely an item is to be correct by chance alone. In turn, this reduces the noise in the measures of metamemory accuracy by reducing the likelihood of chance recognition performance on items that the person does not know" (pp. 465–466). The rationale seems reasonable, but the experiment used only a fixed-number-alternative forced-choice test so the empirical validity of this idea was not, thereby, demonstrated. In an earlier study, Nelson, Leonesio, Landwehr, and Narens (1986) compared a four-alternative to an eight-alternative forced-choice test on what was otherwise the same task. Numerically, the gamma values were higher on the eight-alternative task (.33) than on the four-alternative task (.28). However, the researchers were interested in other effects, and the significance

level of this variable was not explicitly mentioned. So, although it seems reasonable to suppose that the number of test alternatives matters to the accuracy of prediction, we do not know whether this factor really has a detectable effect on the data, and if so, how important it might be. To investigate this question in more detail we conducted a survey of the literature. The results for recognition and for recall are presented below.

Recognition

We were able to find 26 recognition experiments in which the gamma statistic was reported as the measure of metacognitive accuracy. These studies are listed in table 5.1. There were several additional experiments that report other correlation coefficients or methods of expressing the accuracy of metacogniton, but because a number of factors affect these measures differently we decided to restrict the pool to experiments using gamma. The number of alternatives varied from yes/no (which we coded as if it were a two-alternative task) to 19. The left panel of figure 5.1 plots the gamma correlations reported in these studies against the number of alternatives. (The 19-alternative task is not plotted, though those data were entered into the regression equation.) The tasks on which subjects made predictions and were tested included general-information retrieval, picture identification, paired-associate recall, and sentence completion. There was a significant correlation between the number of alternatives and the magnitude of gamma ($r^2 = .21$).

The tasks, in the initial analysis, were rather diverse. It would be expected, then, that factors other than just the number of alternatives would be contributing to the magnitude of the accuracy effects — diluting the correlation between accuracy and the number of alternatives. However, there were 14 experiments in which the same test — general information questions — made up the target material. In most of these experiments, subjects were tested on the same same pool of materials — a set of general information question devised by Nelson and Narens (1980a). Subjects were given a sequence of questions until they had made a certain number of errors. Then they were asked to make FOK judgments on how likely it was that they would later be able to recognize the answer to the question from an

Table 5.1
Accuracy versus number of alternatives in feeling-of-knowing studies

Study	Alternatives[a]	Accuracy[b]	Type of material[c]	Notes[d]
Costermans et al. (1992) exp 1	Yes/no	.11	G.I.	
Butterfield et al. (1988)	Yes/no	.18	Picture identification	Adults
Nelson et al. (1982)	4	.17	P.A.	
Carroll & Simington (1986)	4	.22	P.A.	
Nelson et al. (1984) exp 1	4	.29	G.I.	
Nelson et al. (1984) exp 2	4	.28	G.I.	
Nelson et al. (1986) exp 1	4	.28	G.I.	
Nelson & Narens (1990) exp 1	4	.23	G.I.	
Costermans et al. (1992) exp 2	5	.27	G.I.	
Schwartz & Metcalfe (1992) exp 1, 2	6	.03	P.A.	
Metcalfe et al. (1993) exp 2	6	.18	P.A.	
Prevey et al. (1991)	6	.53	G.I.	Controls
Janowsky et al. (1989) exp 2	7	.42	G.I.	Control
Shimamura & Squire (1986) exp 2a	7	.7	Sentence	Control
Shimamura & Squire (1986) exp 2b	7	.2	Sentence	Delayed
Janowsky et al. (1989) exp 1a	7	.6	Sentence	Controls
Janowsky et al. (1989) exp 1b	7	.5	Sentence	Delayed
Schwartz & Metcalfe (1992) exp 4	8	.27	P.A.	
Metcalfe et al. (1993) exp 1	8	.17	P.A.	

Table 5.1 (continued)

Study	Alternatives[a]	Accuracy[b]	Type of material[c]	Notes[d]
Metcalfe et al. (1993) exp 3	8	.2	P.A.	
Metcalfe (1986) exp 1	8	.45	G.I.	
Nelson et al. (1986) exp 2	8	.33	G.I.	
Nelson & Narens (1990) exp 2	8	.33	G.I.	
Shimamura & Squire (1986) exp 1	8	.50	G.I.	Controls
Schwartz & Metcalfe (1992) exp 3	8	.46	G.I.	
Leonesio & Nelson (1990)	19	.2	P.A.	
Metcalfe et al. (1993) exp 4	Recall	.39	P.A.	
Metcalfe (1986) exp 2	Recall	.52	G.I.	
Jameson et al. (1990)	Recall	.65	G.I.	
Reder & Ritter (1992) exp 1	Recall	.75	Arithmetic	
Reder & Ritter (1992) exp 2	Recall	.76	Arithmetic	

[a]Yes/no indicates yes/no recognition procedure; numbers indicate how many choices given in *N*-alternative forced-choice recognition; recall indicates that recall was final test.
[b]In some cases these numbers are means collapsed across conditions.
[c]G.I. indicates general information questions, P.A. means simple paired associates, others as indicated.
[d]Controls indicate that only gamma correlations for normal control patients are reported.

N-alternative forced-choice recognition test. Subjects were then given the recognition test. Among the 14 experiments there was a reasonable range in the number of alternatives presented. Within this set, there were a variety of differences, including things like delay in testing, other than in just the number of alternatives, but presumably this smaller set was more homogeneous than the total set of 26 in which radically different cognitive tasks and types of materials were included. Therefore we did a second analysis on the general information studies. In this case the correlation between number of alternatives and gamma was sizable ($r^2 = .63$). These data are presented in the center panel of figure 5.1. It seems quite clear from these studies that the number of alternatives is of critical importance to the appearance or nonappearance of a positive FOK-to-knowing correlation.

Recall

If our reasoning, and that of Leonesio and Nelson, about the number of alternatives and the relationship of that variable to observed micropredictive accuracy is correct, then it also follows that the correlations should be higher yet when recall is the second test than when forced-choice recognition is the second test. Recall implies an extremely large number of potential alternatives. We surveyed the literature and found that the correlations are higher in recall than recognition. For example, in one experiment using paired associates (Metcalfe, Schwartz, & Joaquim, 1993), subjects were asked to predict future recognition performance. In an otherwise identical experiment, they were asked to predict future recall performance. Correlations between feeling of knowing and recall (.39) were higher than were those between FOK and recognition (.20). We have reanalyzed these accuracy data (which were not of focal interest in the original article) and found that the difference between these two experiments in gamma correlations was significant, $t(46) = 2.25$, $p < .05$.

There are only five studies in the literature (that we were able to find) that used recall and reported gamma correlations. A mean gamma of .61 was found for those five studies, while a mean gamma of .31 was calculated for the 26 recognition studies shown in the left panel of figure 5.1. This difference in the accuracy of prediction

when the final test was recall rather than recognition was significant, $t(29) = 3.83$, $p < .01$. The recall gammas are plotted in the right panel of figure 5.1.

The results of the review of the literature presented here and of the reanalysis of those data that we had available bearing on the issue are unequivocal in pointing to the importance of the number of test alternatives as a factor determining the magnitude of observed feeling-of-knowing correlations. This is not just a hypothetical confound, which potentially could have some effect. Rather it is a factor that accounts for a considerable proportion of the variability across studies. It seems likely to be a contributing factor leading to some failures to uncover robust and accurate microprediction in humans.

The Hazards of Restricted Range

The restricted range problem occurs when one or both of two to-be-correlated variables represents a restricted or truncated range of the variability of that measure. It results in an observed sample correlation that may be lower than the "real" correlation. If there is little or no difference in the levels found in one variable, then it is impossible to show a correlation with that variable. This is a potential problem with all correlations, and can be acute in studies in which feeling-of-knowing judgments are related to later memory performance. Suppose that the items about which subjects are making FOK judgments are the same in difficulty. Performance scores will tend to cluster around a particular value, and they are therefore restricted. This will obscure a real correlation that might be found in more heterogeneous data. Note that the scores need not be on ceiling or floor for restricted range effects to occur, though these would comprise special cases. When the items are more varied in difficulty, subjects should be better able to discriminate among the items. This should increase the chance of correctly observing accurate judgments given good metacognitive ability on the part of the subject.

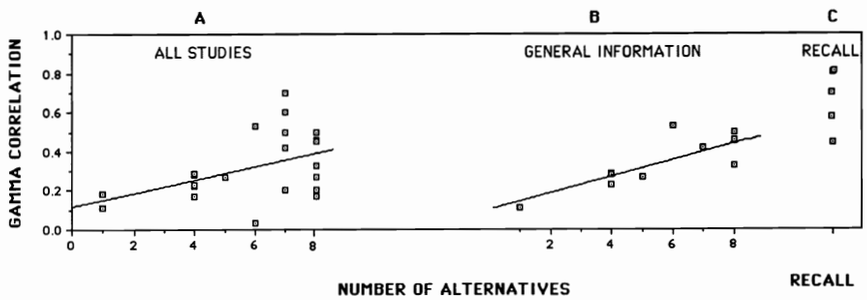


Figure 5.1
Feeling-of-knowing accuracy as a function of the number of alternatives presented in recognition. (A) Plots all studies; (B) plots the general-information studies; (C) shows the five studies using recall as the final test.

Nelson et al. (1986) looked at feeling-of-knowing accuracy when subjects rank ordered a variety of questions relative to each other. In one analysis, they compared the predictive accuracy of two questions that were at the opposite extremes of the ranking (i.e., that enjoyed a considerable range on the scale) to two that were closer together (i.e., that were more restricted in range). The number of intervening questions was important in determining the predictive accuracy for FOK ranking. For questions ranked adjacently (and presumably closest in subjective difficulty), the gamma correlation was only .10, whereas for pairs with six or more intervening items (greater differences in subjective difficulty), the gamma correlation was .45. When they compared only the highest and lowest FOK rank, the gamma improved to .77. Thus it appears that a restricted range on the rating scale, or on the predictive variable, diminishes the correlation observed.

We hypothesized that differences in gamma correlations would also result when the range of item recognition difficulty was varied. To test this idea an experiment was conducted in which two groups of subjects were asked general-information questions with different item-difficulty ranges, as given by Nelson and Narens' (1980a) norms. The restricted variance materials (Narrow) had an initial standard deviation of .18, while the less restricted variance materials (Wide) had a normative standard deviation of .30. They were matched in mean difficulty (.50). Our hypothesis was that correlations giving the accuracy of the judgments would be lower in the Narrow than in the Wide group. Alternatively, though, it was possible that subjects could adjust their level of "tuning" of the judgments to

the level of variance in the material, and no differences would be seen.

Subjects read the general-information questions, answered the questions aloud, and an experimenter recorded their responses, until each subject had made 10 errors. Then the 10 cards were reshuffled and subjects were requested to give FOK judgments, on a 0 to 10 scale, about the likelihood that they would later be able to choose the correct answer from among eight alternatives. Then, the recognition test, using the materials of Wilkinson and Nelson (1984), was conducted.

Because subjects only made judgments on the unrecalled items, whereas the norms included all items in the set, we checked to see that the means and variances maintained the same relation for the unrecalled items as they did the original pool in both conditions. The two conditions maintained equivalent mean difficulty for nonrecalled items (Narrow = .24, Wide = .21, $t(38) = 1.40$, n.s.). The standard deviations — giving the manipulated difference in range that was the main experimental variable of concern — were still significantly different, in keeping with the initial manipulation (Narrow = .07, Wide = .24, $t(38) = 8.16$, $p < .05$).

The mean gamma correlation for the Wide condition (.72) was higher than that of the Narrow condition (.43), $t(38) = 3.02$, $p < .05$. Both correlations were significantly greater than zero (i.e., better than chance). The conclusion is straightforward: Restricted range effects can substantially alter the magnitude of an observed correlation, and might obscure a real correlation. This danger is especially important when comparisons across different treatment combinations or different subject populations are of interest in the experiment. If a researcher is investigating the possibility that there are different cognitive operations implied by a difference in the correlations across tasks (see Metcalfe, 1986a, for example) or that different patient groups exhibit selective deficits in a particular kind of ability, as measured by a difference in the correlations (see Shimamura & Squire, 1986b; Janowsky, Shimamura, & Squire, 1989, for example) it is of considerable importance to ensure that the groups being compared are using the same range of scale on the judgments task, and also that they exhibit the same magnitude of variability on the criterion (recognition) task. Otherwise, the observed differences

may be attributable to restricted range effects rather than to the more interesting cognitive variables under study.

Different Levels of Performance on Criterion Tasks: Dissociation Studies

Accuracy differences in metacognitive judgments across patient groups with different focal lesions or with different neuropsychological syndromes could be of enormous importance for our understanding of the structure and function of the systems and the interrelation among the systems involved in human cognition. A number of studies have been directed at investigating such differences in underlying systems or processes (Janowsky et al., 1989; Lupker, Harbluk, & Patrick., 1991; Metcalfe, 1986a; Nelson, Leonesio, Shimamura, Landwehr, & Narens, 1982; Prevey, Delaney, Mattson, & Tice, 1991; Shimamura & Squire, 1986b). In this section we discuss problems (and methods of solution to those problems) that, while not necessarily isolated to these kinds of studies, nevertheless are salient in them. The possibility of restricted range confounds, as discussed above, is, of course, one such possible pitfall.

A second possible confound, noted by Shimamura and Squire (1986b), is an item selection effect. If, for some reason, one group of subjects ends up with a set of items that are easier or more difficult than another group, differences between the two groups might be due to item differences, rather than to differences in the underlying cognitive structures of the two groups. Shimamura and Squire (1986b) investigated whether certain kinds of amnesic patients would show selective deficits in metacognition, that could be teased apart from general impairments in memory. In experiment 1, Korsakoff amnesics, amnesics of varied etiologies other than Korsakoff's syndrome, and control patients were presented with general-information questions and asked to recall them. Both control patients and the non-Korsakoff amnesics showed normal above-chance micropredictions. By contrast, the Korsakoff patients were impaired. Unfortunately, they were also impaired relative to both controls and other amnesics in recall of the general information. Presumably that information was learned before the onset of the amnesia, and hence was somewhat spared from the memory impairment. As a result of

the differences in recall, though, the pool of questions the Korsakoff patients failed to answer, and hence the questions that they made judgments on, was on average easier than that of the other subjects, since the Korsakoff patients did not skim off the easy questions by recalling them in the first phase of the experiment. Shimamura and Squire (1986b) suggested that the decreased accuracy seen in the Korsakoff patients might have been due to a specific metacognitive deficit for Korsakoff patients, but alternatively that it could have resulted because of differences in the question pools.

Accordingly, in experiment 2 they used questions that had to be newly learned in a successful effort to equate recall on the initial test among the different amnesics. "Facts" such as "At the museum, we saw ancient relics made of clay" were used. At test, subjects were shown the sentence with a word missing: "At the museum, we saw ancient relics made of _____." Metacognitive accuracy was still impaired only for the Korsakoffs patients. There are really two possible factors that could have been contributing to Shimamura and Squire's results, other than the factor of interest. One is an item selection effect. The other is simply that metacognition might necessarily be correlated with goodness of memory (or indeed, they might be causally connected). By equating the initial level of recall, though, it is difficult to argue that either of these effects was responsible for the metacognitive differences seen. This leads to the interesting conclusion that Korsakoff patients suffer a real impairment in metacognition, dissociated from other cognitive operations. This finding has been followed up by Janowsky et al. (1989) with frontal patients, and has far reaching theoretical implications for the relation between metacognitive and control processes and basic memory (Metcalfe, 1993).

Given that different patient groups may have different base levels of final recognition, as well as initial recall, it is important that the statistic used to measure metacognition not be sensitive to differences in the absolute level of recall. The nonparametric gamma correlation measure, preferred by many researchers, has the desirable characteristic of being independent of absolute performance in the final test. Table 5.2 gives several examples of patterns of data that could easily result were the baseline performance across subject groups different. Intuitively one would say that the correlation should be

Table 5.2.
Hypothetical cases demonstrating differences between r and g[a]

Case 1		Case 2		Case 3	
Judgment	Recognition	Judgment	Recognition	Judgment	Recognition
100	C	100	C	100	C
90	I	60	C	90	I
90	I	50	C	90	I
90	I	40	C	90	I
90	I	30	C	90	I
90	I	20	I	10	I
$G = 1.00$		$G = 1.00$		$G = 1.00$	
$r = 1.00$		$r = 0.52$		$r = 0.32$	

[a] C indicates correct response, I indicates incorrect response, G indicates Goodman–Kruskal gamma correlation, r indicates Pearson product–moment correlation.

perfect in all three cases. The gamma correlation exhibits such a perfect correlation, whereas the parametric Pearson r shows differences. On the other hand, the gamma correlation obscures magnitude information that in some cases could be revealing real judgmental differences, so some caution is needed in its use. A subject might be revealing something real and interesting about the set he or she is judging and/or about the judgment processes themselves by giving values of .49, .50, and .51 for the three items in one list and .01, .50, and .99 in another. But the gamma statistic, treating only rank, discards this information. Nelson (1984) argued that gamma is preferable to parametric signal detection theory analysis because it does not require the data to be normally distributed, and is preferable to nonparametric signal detection analysis because gamma does not require as many observations. However, Swets (1986) showed that for 2×2 contingency tables, that gamma (identical in this case to Yule's Q) yields equivalent outcomes to signal detection analysis.

As described above, Shimamura and Squire (1986b) were concerned that an item selection effect might somehow lead to very low predictive accuracy in a low-functioning group. But there is a reverse side to the item selection that leads us to suppose that just the reverse would be expected to occur. Consider a situation in which an ex-

tremely bright and well-informed subject participates in a general information study and answers most of the questions in the pool. It may be surprising, at first, that such a subject does not invariably show good micropredictive accuracy, since we tend to expect that good performance will be related to good metacognitive ability (though see the section below on skills for some counter examples). The reason for poor microprediction given exceptional memory performance is straightforward, and combines item selection effects with restricted range effects. The extremely good subject necessarily limits the pool of items to those that are very difficult, and hence operates within a restricted range. The more mediocre subject is tested on a broader mix of easy and difficult questions. As we showed above, the chance of revealing a high correlation between prediction and performance decreases under restricted range conditions. Thus, because of self-imposed item selection effects, the extremely good subject is operating within a restricted range of question difficulty and is liable to show worse micropredictive accuracy than the more mediocre subject. This effect should occur at all levels of performance with the better subjects showing worse micropredictive accuracy than the poor ones, all else being equal.

Nature of the Test: Subjects' Expectations and Subjects' Skills

Knowledge of the Test

One might hope that some metacognitive judgments or feelings reflect a stable familiarity with or propensity toward certain items — a propensity that would have predictive accuracy about performance on a variety of tasks. Unfortunately, a number of studies suggest that the tasks and the metacognitive judgments are much more specific. If the test is different from that which the subject expected when making the judgment, the correlation between the judgment and test, whether positive, negative, or zero, is suspect. It follows that if subjects do not have a full appreciation for what the final test will be, or if they can enact the test by using a different strategy than that implied as being necessary when they were making the judgment, then inferences about the goodness of metacognition with respect to that test cannot be made. For example, in one experiment investigating the relation between feeling-of-knowing judgments, recognition, and perceptual identification, Nelson, Gerler, and Narens (1984) asked subjects to provide FOK judgments about later recognition. The results showed that these judgments did, modestly, predict recognition, gamma = .29, and that they also predicted perceptual identification, gamma = −.16. (The negative correlation resulted because a low rather than a high number of perceptual recognition trials implied good performance.) The absolute values of the correlations were not significantly different from each other. Unfortunately for the stable propensity idea, there was no correlation between the absolute values of the gammas relating the FOK judgments and recognition and the gammas relating the FOK judgments to perceptual identification, based on these scores for each subject. So one is not justified in concluding that the FOK judgments are based on simple strength of memory trace and that that strength influences both recognition and perceptual identification in the same way. It is not clear what one should conclude. As Nelson et al. (1984) discussed, though, there does not seem to be a stable unidimensional FOK ability that applies to all tests.

Sometimes experimenters inadvertently construct a test situation in which the subject can make the response in a manner other than that which is expected when they are making the judgment. For example, in one experiment in Metcalfe et al. (1993) subjects were given a list of paired associates to study. They were asked to give judgments about the likelihood that they would be able to choose the correct associate to the cues. However, only one of the alternatives presented at test (the correct one) had been presented in the initial list. To do the task, subjects needed only to assess the familiarity or oldness of the alternatives; no associative knowledge was necessary. As has been pointed out by Blake (1973) the inclusion of other old items among the lures is needed to make the test one of association, rather than just of familiarity. Some of the other experiments in the Metcalfe et al. (1993) study included within-list lures. Unfortunately, in some conditions the gammas were better with intralist lures, whereas in other conditions they were worse, so a clear systematic factor cannot be isolated.

Glenberg and his colleagues examined how knowledge about the structure of the test affects accuracy of judgments (Epstein, Glen-

berg, & Bradley, 1984; Glenberg, Wilkinson, & Epstein, 1982; Glenberg & Epstein, 1985, 1987; Glenberg, Sanocki, Epstein, & Morris, 1987; Morris, 1990). Their interest was in how well people understood text they had just read, and thus, they asked subjects to make judgments of comprehension. These were correlated then with performance on various objective tests of comprehension.

Glenberg et al. (1982) and Epstein et al. (1984) investigated the relation between self-assessment of comprehension and objective performance on a contradiction detection task. They asked subjects to read short passages of text as if they were preparing for an exam. Subjects were also informed that inconsistencies in the text could occur, and that it was their job to detect these contradictions. They were instructed to indicate contradictions in the text, and then rated their confidence that they understood the texts. Presumably, if subjects understood the text they should be able to pick out the inconsistencies. The results showed that subjects frequently reported high confidence of comprehension, but nevertheless failed to detect the contradictions contained in the passages. Glenberg et al. (1982) labeled this phenomenon an "illusion of knowing."

Similarly, Glenberg and Epstein (1985) asked subjects to read a brief passage of text and to judge their confidence in being able to correctly draw inferences from that text (on a 1–6 scale). They were then asked to verify whether inferences drawn from the text were true or false. The inferences were new statements whose truth value could be determined by the information provided in the text. For example, after reading a passage entitled the "Detection of Black Holes" and making a confidence judgment, subjects were given one of two sentences, "A black hole which exists in the region of the universe many light years removed from any other matter has a high probability of going undetected" (true), or "A black hole which exists in a region of the universe many light years removed from any other matter has a low probability of going undetected" (false). Then they indicated whether they thought their choice was correct or incorrect.

In the first two experiments using this procedure, the judgments of comprehension failed to predict performance on the inference verification task (but see Weaver, 1990, for better accuracy in a similar paradigm). Subjects were, however, able to assess the correctness of

their answers, after the fact. It is possible that subjects were simply unable to assess their comprehension. The third experiment, in the series, however, suggests a different possibility.

In the third experiment, Glenberg and Epstein (1985) modified their procedure by adding an additional confidence judgment and inference verification. Subjects judged whether they would be correct on the objective inference test, were given a test question, and judged the correctness of their answer (so far identical to the first experiments). Then they made a second confidence judgment about how likely they would be to correctly answer a second objective inference test. They then answered the question and judged the correctness of their second answer. On second metacognitive prediction, subjects showed a modest ability to predict their performance. Apparently, the reason that subjects were able to accurately judge their performance on the second question but not the first lies in the fact that experience with the first question gave them a feel for the nature of the test. They were, by and large, correct in saying whether they got the answer correct on the first question, and so had some (better) basis for saying whether they would get a similar second question correct.

Glenberg et al. (1987) found a similar pattern for other kinds of judgments. In a series of experiments, poor predictive accuracy resulted in tests of verbatim recognition, idea recognition, and inference-making, regardless of whether the tests were given immediately or after a delay. But, when subjects were given a second test, predictive accuracy improved. Glenberg et al. (1987) suggested that the first test provided the subject with needed feedback on the nature of test, as well as feedback on his or her actual comprehension of the text. Knowing the structure of the test, subjects were in a better position to make accurate metacognitive judgments.

Skill or Knowledge in the Domain

There are those who might say that the relation between people's skill and their metacognitive ability should not be treated as a methodological problem, since skill is a psychological/cognitive variable of great interest in its own right. While we certainly agree with this

sentiment, we could not resist the temptation to include a brief section on skill. Our rationalization for doing so is that as a person gains skill in a domain he or she may have a better knowledge of the nature and difficulty of the test. Thus skill may be seen as a corollary of the idea that, in order for a fair assessment of metacognitive ability to be possible, the subject must have some idea of what the structure of the test questions might be. Of course, other arguments for increases in metacognitive performance with skill can be made, and we do not wish to discount their importance. For example, people may become highly skilled and able to cope with problems in a particular domain because they have good metacognitive abilities (rather than the reverse). In short, then, when compared to more straightforward and clearly methodological issues like the number of alternatives at test, or restricted range effects, including "skill" as a methodological confound is borderline.

The idea that increasing experience, skill, or knowledge within a particular domain should result in increasingly accurate metacognitive judgments seems intuitively obvious. Some studies reveal an effects of skill. Maki and Berry (1984) asked subjects to read sections out of an introductory psychology text. They then made judgments as to how well they thought they would do on multiple-choice questions for that text. In one condition, subjects returned 24 hours later and were given a four-alternative choice test. Those subjects who performed above the median on the test showed good predictive ability; those below the median showed no predictive ability. In addition, the studies of Glenberg et al. (1987) and Glenberg and Epstein (1985) discussed in the preceding section, in which subjects' performance improved with self-generated feedback, or with additional testing, could be considered to be due to skill.

On the other hand, Nelson and Narens (1990) found no improvement in the accuracy of FOK judgments when subjects were given feedback on an earlier session. Subjects were informed of what their judgment had been and of whether they had given the correct answer in recognition. But this feedback had no effect. Similarly, in one experiment reported in Metcalfe et al. (1993) subjects participated in four sessions of learning, initial recall, FOK judgment, and final recognition. The sessions were structurally identical. There was no improvement in accuracy over sessions.

Glenberg and Epstein (1987) examined the relation between judgments of comprehension and inference verification in a group of music or physics "experts" (students who had completed two or more courses in at least one of the topics). Subjects read a number of passages concerning both music and physics, made judgments about how well they understood the passage, and then were given sentences to verify. The physics students gave higher judgments to the physics than the music passages and tended to be more correct on the physics passages. The reverse was found for the music students. Thus both groups showed reasonable macropredictions. However, none of the students was able to predict which passages would be easiest, either within their own speciality domain or in the other domain. Thus the micropredictions were near chance. Glenberg and Epstein (1987) suggested that subjects use a system of self-classification (as expert or nonexpert) to determine their overall confidence. This accounts for both the good accuracy across domains, although the failure to find discrimination is perhaps surprising. It might be noted, however, that the criterion test was a simple true/false test, so there were few alternatives, and a low predictive accuracy might have resulted for that reason alone, as illustrated in the first section of this chapter. Alternatively, the effect of skill on metacognition might be overrated.

Despite the negative results of Glenberg and Epstein (1987), Nelson and Narens (1990), and Metcalfe et al. (1993), discussions of the nature of the novice/expert shift in problem solving (see, for example, Wiser & Carey, 1983) would lead one to suspect that if novices were requested to give rankings of problem difficulty they would be rather poor at doing so. Experts, though, understanding the nature of the concepts underlying various problems, should be better able to assess problem difficulty. As before, though, the exact nature of the test, and the relevance of the skill possessed by the expert and not possessed by the novice, is likely to be critical.

An experiment conducted by J. Krause, in the second author's laboratory, provides a clear example of how skill or knowledge in the task domain may impact on metacognitive judgments. Krause constructed a set of crypt arithmetic problems somewhat similar to the famous DONALD + GERALD = ROBERT problem. The subject's task was to assign a digit to each letter, such that the arithmetic

worked out correctly. The twist in Krause's experiment was that he constructed the easiest problems to look like they were long and complicated, and the difficult problems to appear perceptually simple. So, for example, the problem:

```
    H H H H H H H H H H
+ S A A A A A A A A A S
  R A A A A A A A A A A
```

looks fairly difficult, but in fact is easy.[1] Krause constructed, and subjects were tested on, two sets of such problems that varied in their real difficulty in a way that, in his subjective impression, was negatively correlated with their perceptual complexity. The structures of the problems in the two sets were identical. What varied were the actual letters that each problem was assigned. Before solving, subjects were asked to make judgments about how difficult each of the problems would be, as measured by how long it would take to solve it. These predictions were then correlated with the actual time to solve. The difference in the accuracy of the predictions from the first to second trial was dramatic. Correlations, giving subjects' micropredictive accuracy, were near zero on the first trial and near perfect on the second. Of course, this was an extreme situation, where the rules and difficulties of each problem were easily and immediately assimilated during the solving process. The requisite skill needed to understand the structure of the crypt arithmetic problems presented was gained in a single trial, with a corresponding increase in metacognitive "ability." Despite the transparency of this situation, it provides a demonstration of the kind of increase in metacognitive judgment one might expect to accompany an increase in skill in other well-defined situations.

It appears likely that the nature of the task may interact with the efficacy of feedback or skill in improving predictive judgments. In some situations subjects may learn the structure of the task. When they do so their predictive ability may improve dramatically. In other situations, however, such as memory predictions, the structure of the final test may be less learnable, and the improvement in predictive ability with experience in the task less obvious.

Conclusion

The study of human metacognition and its relation to other cognitive processes seems fraught with technical and methodological difficulties. We think that these problems are solvable and that, with careful experimental and theoretical techniques, valuable and valid results can be attained that will advance our understanding of human cognition.

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Note

1. The solution is:

```
    9 9 9 9 9 9 9 9 9
+ 1 0 0 0 0 0 0 0 0 1
  2 0 0 0 0 0 0 0 0 0
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